

University of Colombo School of Computing

SCS 3312 : Research Methods

D3: Data Summary and Preliminary Findings

**“An Investigation into Student Behavior During Free Periods
Between Scheduled Lectures at UCSC”**

Student Names and Indexes

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We are also thankful to the sixty UCSC undergraduate students who gave their time to complete the survey instrument at the five field sites, and whose honest responses made this quantitative phase possible.

Abstract

University students have gaps between lectures, but little is known about how they actually spend this time or whether campus spaces meet their needs. Understanding this helps the university plan better spaces and support student wellbeing. This report is based on the D1 fieldwork and the survey built in D2. The final 19-question survey was given on paper to 60 students across five campus locations: Library, Canteen, Dewale, Common Room, and Open Area.

The collected results were analysed using frequencies, percentages, averages, and simple comparisons. Results are organised into five sections: participant background, factors affecting location choice, space usage and avoidance, free-period activities, and overall satisfaction with campus space.

The results confirmed several patterns from D1: free-period length strongly affects both what students do and where they go, the Canteen and Open Area are avoided mainly for environmental reasons, and many fourth-years spend their free time on research supervision rather than at the five main locations. One surprising result was that third-year students, despite having full access to all spaces, reported the lowest satisfaction overall.

These findings, along with a critical analysis of the study's limitations, matter because they turn D1's early observations into evidence backed by a larger sample, and they will be combined with the D1 findings in the final D4 report.

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List of Acronyms

UCSC	University of Colombo School of Computing
SD	Standard Deviation
N	Total number of respondents

1 Introduction

According to the D0 proposal and the D1 fieldwork, UCSC students spend a large amount of time in free periods between lectures. Most students use one of five main campus locations during free periods: Library, Canteen, Dewale, Common Room, and Open Area.

In the D2 report, 19 survey questions were developed covering five main sections. The survey was tested with 10 students and several improvements were made before the final version was used.

This D3 report explains how the full survey was carried out with 60 students and presents the results using descriptive statistics. Chapter 2 covers how the data was collected and cleaned, including an important decision on how to handle access restrictions found in the data. Chapter 3 presents the statistics for each survey section, and Chapter 4 looks at cross-cutting patterns. Chapter 5 compares these results against the five cross-location insights from D1 and the six research objectives from D2, checking which findings are confirmed, only partly confirmed, or not tested by the survey. Chapter 6 closes with a critical analysis of the study, and Chapter 7 presents the conclusions, setting up the integration of these results with the D1 qualitative findings in the final D4 report.

2 Data Collection Summary and Data Quality

2.1 Execution of the Data Collection Plan

The finalised survey instrument (D2, Chapter 6) was printed and distributed by the research team to students at the five selected field sites during free-period hours, following the stratified convenience sampling method described in the D2 data collection plan. A total of 60 survey forms were distributed, completed by the participants, and collected by the research team. The responses were then entered into a structured Microsoft Excel workbook by the three group members for analysis. This met the target sample size of 60 respondents. Table 2.1 summarises the achieved sample.

As explained in Section 2.2, the “Common Room” field site refers to two separate year-specific common rooms. However, it is treated as a single field site throughout this report to maintain consistency with D0–D2, rather than being presented as six separate sites.

Parameter	Value
Total valid responses entered	60
Data collection period	2 July – 10 July 2026
Administration mode	Printed survey form, self-completed on site
Distribution	Convenience sampling across the five field sites

Table 2.1: Achieved sample characteristics ($N = 60$)

2.2 Handling Structurally Restricted Spaces

The D1 fieldwork found that the Dewale is only accessible to second-year students and above, meaning first-year students cannot enter it. However, the D0, D1, and D2 reports all referred to the Common Room as a single location without explaining that UCSC actually has separate common rooms for second-year and third-year students. First-year and fourth-year students do not have a dedicated common room.

The D1 fieldwork team observed and physically described only the third-year common room; the second-year common room was deliberately not visited, because the presence of third-year researchers could have influenced the behaviour of students using that room (an observer effect). This decision was not explicitly explained in the D1 report.

The D2/D3 questionnaire also listed “Common Room” as a single option in Questions B1, B2, and B3, without specifying which common room it referred to. Therefore, this report assumes

that respondents answered based on the common room available to their own year group. This interpretation is supported by the survey results: none of the 21 second-year respondents reported that the Common Room was inaccessible, suggesting that they understood the question to refer to their own second-year common room rather than the third-year room observed during D1.

So, the genuine lack of provision applies only to first-years, who are restricted from the Dewale and do not have a separate common room, and to fourth-years, who also do not have a separate common room – so their reported lack of access represents a real limitation.

Among the 34 respondents who said they actively avoided at least one campus space (Question B2), 20 mentioned at least one preference-based reason, such as noise, unpleasant smells, crowding, or distance, while 18 respondents reported avoiding a space because they were not allowed to access it. Therefore, this report analyses structural restrictions separately from preference-based avoidance, as combining them would misrepresent genuine access limitations as simple user preferences.

3 Descriptive Statistics and Findings by Section

3.1 Section A: Participant Background

Table 3.1 and Figures 3.1–3.2 summarise the background profile of the 60 respondents.

Variable	Category	n (%)
A1. Year of study	1st Year	14 (23.3%)
	2nd Year	21 (35.0%)
	3rd Year	21 (35.0%)
	4th Year	4 (6.7%)
A2. Free period duration	Less than 30 min	5 (8.3%)
	30–60 min	18 (30.0%)
	1–2 hrs	20 (33.3%)
	More than 2 hrs	17 (28.3%)
A3. Free periods per week	0–1	20 (33.3%)
	2–3	22 (36.7%)
	4–5	6 (10.0%)
	6 or more	10 (16.7%)

Table 3.1: Section A: Participant background ($N = 60$)

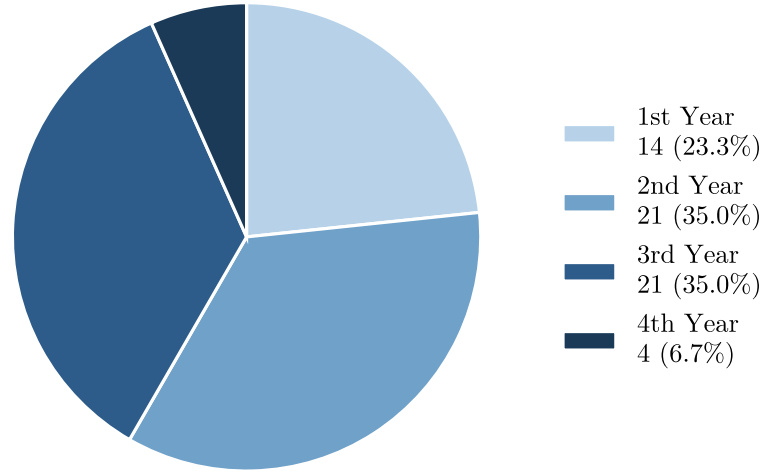


Figure 3.1: Respondents by year of study

Second- and third-year students dominate the sample (35% each), while fourth-years make up only 6.7%. This uneven spread was anticipated in the D1 report, which recorded fourth-year students explaining in their own words that they are rarely present at the five surveyed locations because their free time is absorbed by final-year research work and supervisor meetings – a pattern this dataset independently reproduces (Section 3.4).

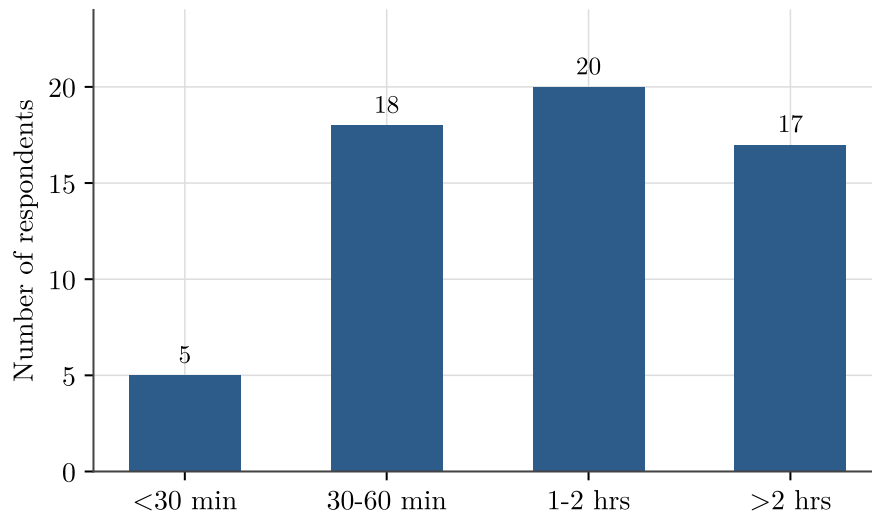


Figure 3.2: Typical free period duration

The chart shows how long students’ typical free periods last, across all 60 respondents. Most students fall into the two mid-to-long duration bands: 20 students have 1–2 hour gaps and 18 have 30–60 minute gaps, while 17 report gaps longer than 2 hours. Very short free periods (under 30 minutes) are the least common, reported by only 5 respondents.

3.2 Section B: Space Usage and Avoidance

Most frequently visited locations (B1)

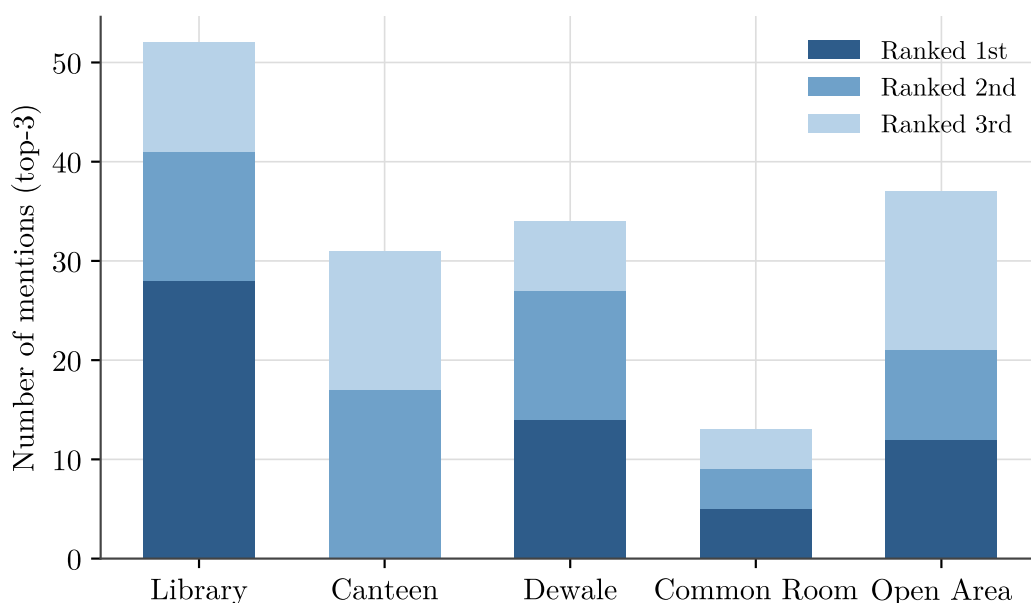


Figure 3.3: Top 3 most frequently visited locations, by rank

The Library is the clear leader as students’ single most-visited location, named rank 1 by 46.7% of respondents (28/60) and appearing somewhere in the top-3 of 52/60 respondents (86.7%). The Dewale (23.3% rank-1) and Open Area (20.0% rank-1) follow. Notably, the Canteen was never ranked as the single most-visited location by any respondent, yet it appears in 31/60 (51.7%) top-3 lists, almost always as a 2nd or 3rd choice. This is not a sign that the Canteen is unimportant – it is evidence that it functions as a supporting stop within a free period rather than the reason for it, which fits the D1 finding that food and socialising happen alongside, not instead of, a student’s main activity. The Common Room’s low totals across all three ranks (13 total mentions) point to a smaller, more habitual user base rather than broad appeal.

Space avoidance (B2 and B3)

34 of 60 respondents (56.7%) reported actively avoiding at least one campus space. Following the decision explained in Section 2.2, avoidance reasons are reported in two separate figures.

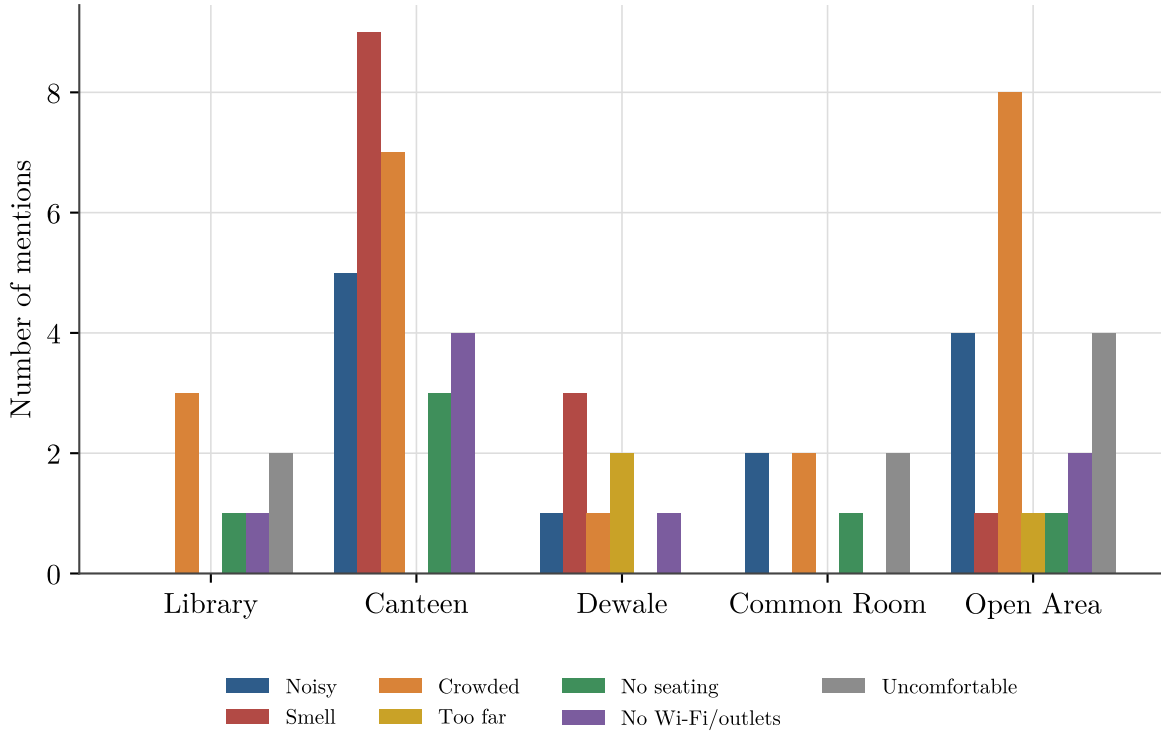


Figure 3.4: Genuine preference-based avoidance reasons by location

Figure 3.4 does not just tell us how many students avoid each location – it independently confirms what the D1 interviews and observations already surfaced. The Canteen’s top reason, unpleasant smell ($n = 9$), directly echoes the unprompted D1 finding that the Canteen was avoided because of odour and flies. The observation that the Canteen becomes overcrowded at peak times is also confirmed, since many students cited “being crowded” as a reason for avoiding it. Similarly, the main reason for avoiding the Open Area is being over-crowded, and its high “uncomfortable” count lines up with D1’s observation that the Open Area was avoided for being too hot. The Library’s small but real cluster of “uncomfortable” citations ($n = 2$) is also consistent with D1’s divided-response finding, where one interviewed student described the library as feeling cramped even as others praised it for being calm and quiet. The one departure from D1 is the Dewale’s three “unpleasant smell” citations – a reason D1’s fieldwork never flagged for that location at all, suggesting this may be a genuinely new finding to investigate.

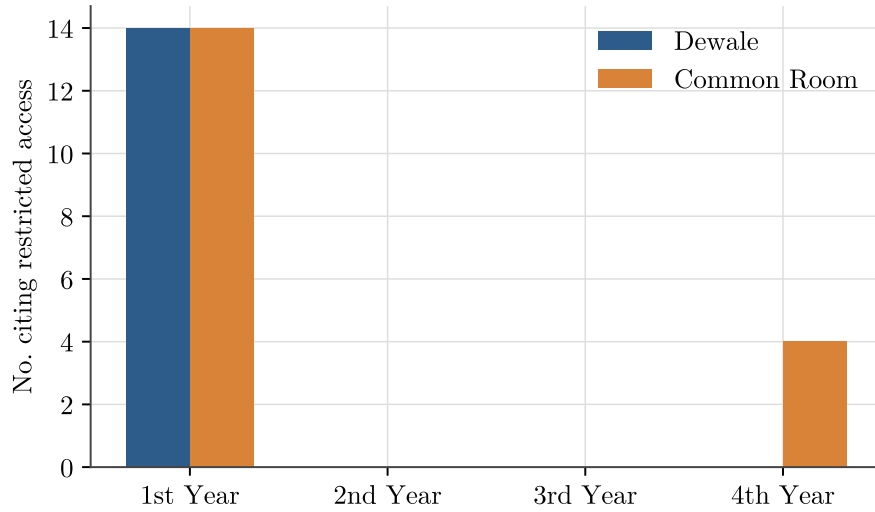


Figure 3.5: Structural access restriction cited as an avoidance reason, by year of study

All 14 first-year respondents cited the Dewale as restricted, and the same 14 first-years plus all 4 fourth-years cited the Common Room as restricted – reflecting a genuine provision gap, since neither year group has a dedicated common room. No second- or third-year respondent reported either restriction, consistent with each having their own accessible space (as discussed in Section 2.2). The genuine access gap is therefore confined specifically to first-years and fourth-years.

3.3 Section C: Factors Influencing Location Choice

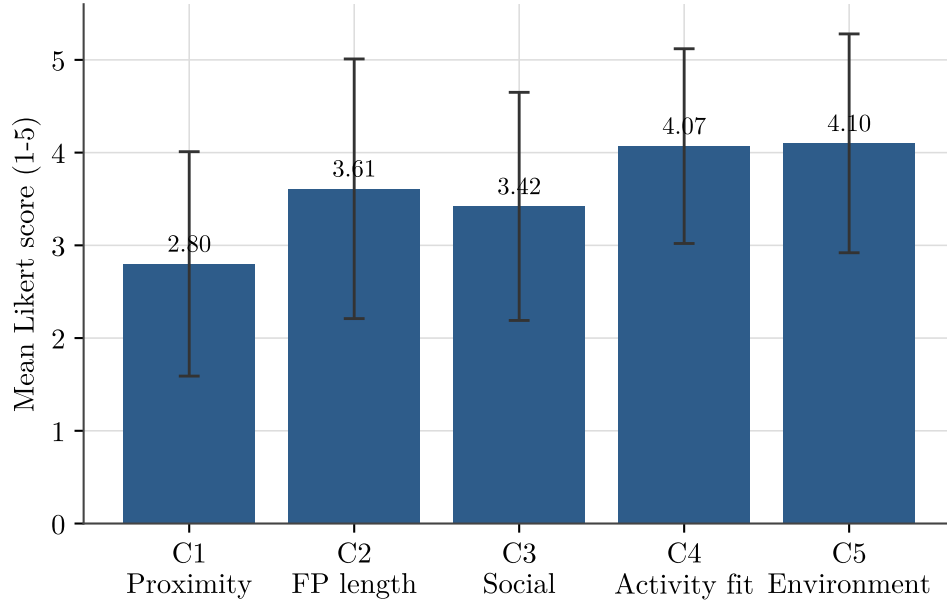


Figure 3.6: Mean agreement with Section C statements (error bars = 1 SD)

Item	Statement	Mean	SD
C1	How often does the location of your next lecture influence where you spend your free period?	2.80	1.21
C2	The length of my free period affects where I choose to spend it.	3.61	1.40
C3	I choose locations where I am likely to find friends or classmates.	3.42	1.23
C4	I prefer locations that match the activity I plan to do (e.g., studying, resting).	4.07	1.05
C5	Environmental conditions (e.g., heat, noise) influence my choice of location during free periods.	4.10	1.18

Table 3.2: Section C: Likert item means and standard deviations (1–5 scale, $N = 60$)

The results show that environmental conditions (C5, mean = 4.10) and the suitability of a location for the intended activity (C4, mean = 4.07) were the two most important factors

influencing students' choice of free-period spaces. This directly supports the D1 themes that environmental conditions filter who uses each space and that students match location to activity rather than habit.

Although proximity to the next lecture (C1, mean = 2.80) had the lowest overall mean, its importance depends on the length of the free period. It becomes more important during short breaks but less important during longer ones, so the overall average does not fully reflect its influence (this can be seen clearly in Chapter 4's duration-vs-primary-activity analysis).

Free-period length (C2, mean = 3.61) showed the greatest variation in responses, suggesting that students differ considerably in how much this factor affects their decisions. This may be related to differences in the duration of their own free periods.

Social motivation (C3, mean = 3.42) was a moderately important factor, indicating that spending time with friends influences location choice (which was also observed in the Open Area and Canteen during D1 observations) but is generally less important than environmental conditions or activity suitability.

3.4 Section D: Activities During Free Periods

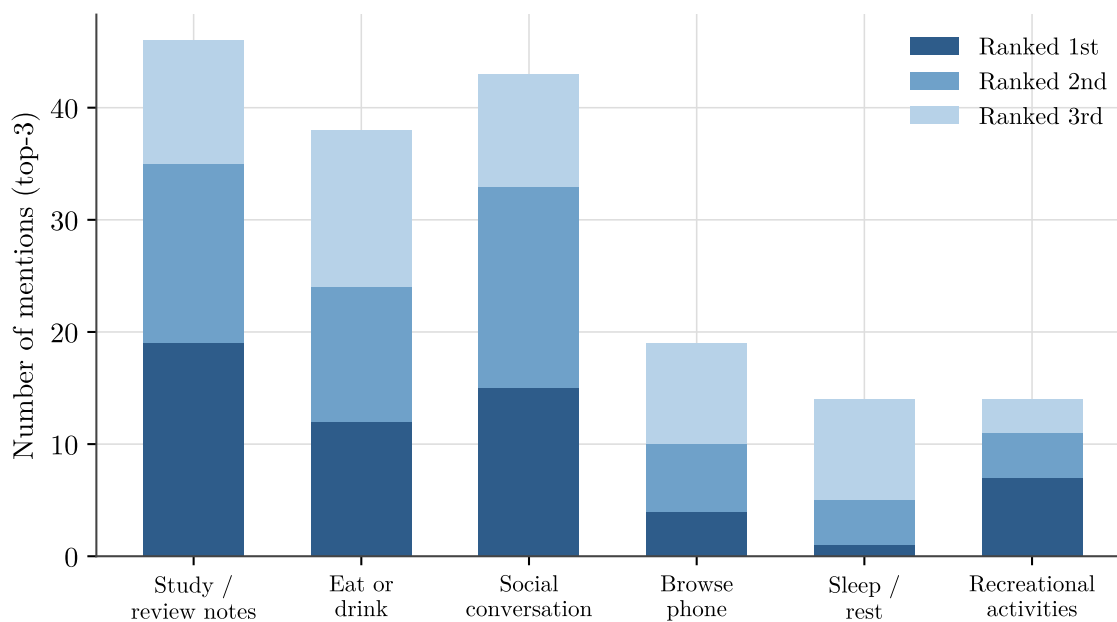


Figure 3.7: Top-3 most common free-period activities, by rank

Studying or reviewing notes is the most common single (rank-1) activity (31.7%, 19/60), followed by social conversation (25.0%) and eating or drinking (20.0%). Passive phone browsing was ranked 1st by only 4 respondents (6.7%) but appeared somewhere in 19/60 (31.7%) top-3 lists, indicating it is widespread as a secondary, fallback activity rather than

a dominant primary one – a useful refinement of the D1 claim that phone browsing is the “default fallback activity”. Four respondents (all fourth-years) wrote in “meet/attend project supervisors” as an unlisted activity, independently confirming that fourth-year free time is absorbed by research supervision rather than the activities listed on the survey.

Perceived value and frequency of phone browsing (D2 and D3)

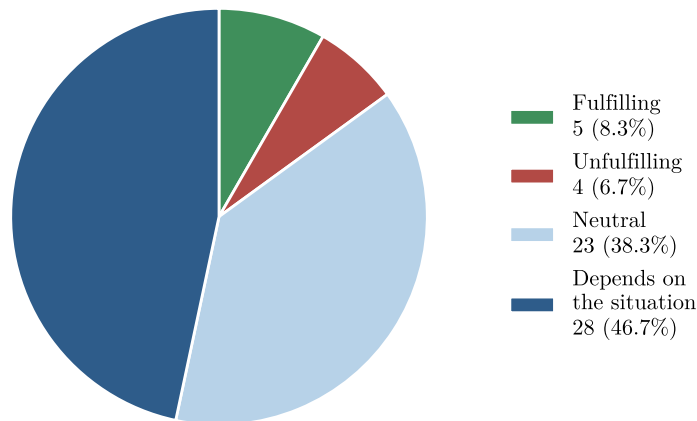


Figure 3.8: Perceived value of phone browsing

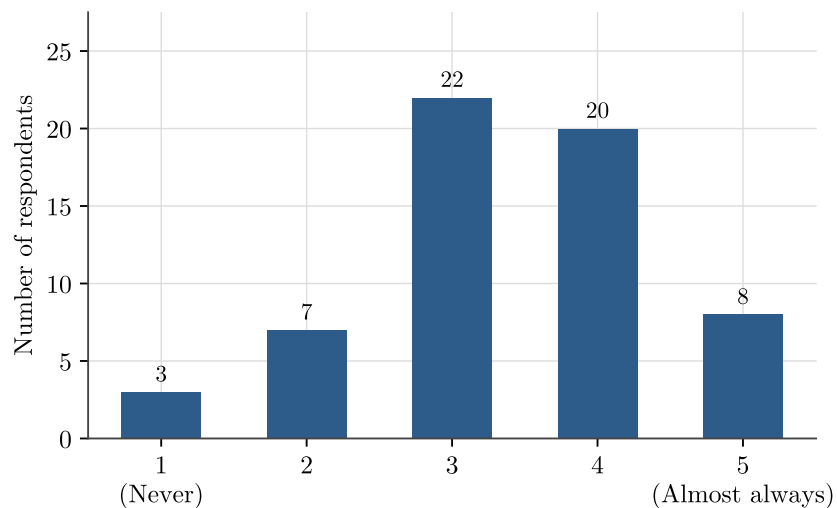


Figure 3.9: Distribution of D2 (frequency of passive phone browsing) – Mean = 3.38, SD = 1.03

The results show that 85% of students picked one of the two “safe” answers (Neutral or

Depends), and only a small number chose the negative option (6.7%), meaning students are still avoiding a direct judgement on their own phone use rather than genuinely being neutral about it. This fits the wider pattern seen elsewhere in the data: phone browsing is common but never a main activity (only 4 of 60 ranked it first), it shows up mostly as a secondary, in-between activity (31.7% mention it in their top three), and D1’s fieldwork already noted that how someone uses their phone depends heavily on the situation (waiting in a queue versus scrolling alone). Put together, students seem to browse their phones often, but are consistently reluctant to say clearly whether they think it is a good use of their time.

Spending free periods with friends (D4 and D5)

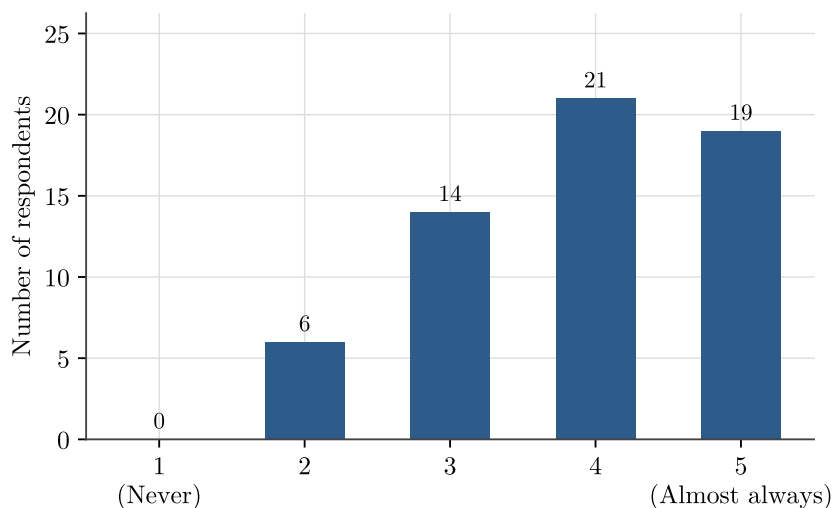


Figure 3.10: Frequency of spending free periods with friends – Mean = 3.88, SD = 0.98, N = 60

Spending free periods with others is the common pattern among students. No respondent selected “Never”, and 40 out of 60 respondents (66.7%) reported that they often or almost always spend their free periods with others. This supports the D1 field observations, which found that students commonly gathered in pairs or small groups in locations such as the Open Area, Canteen, and Library. These results also provide a clear quantitative baseline for the D2 research objective, showing that group use is the dominant pattern while spending free periods alone is less common.

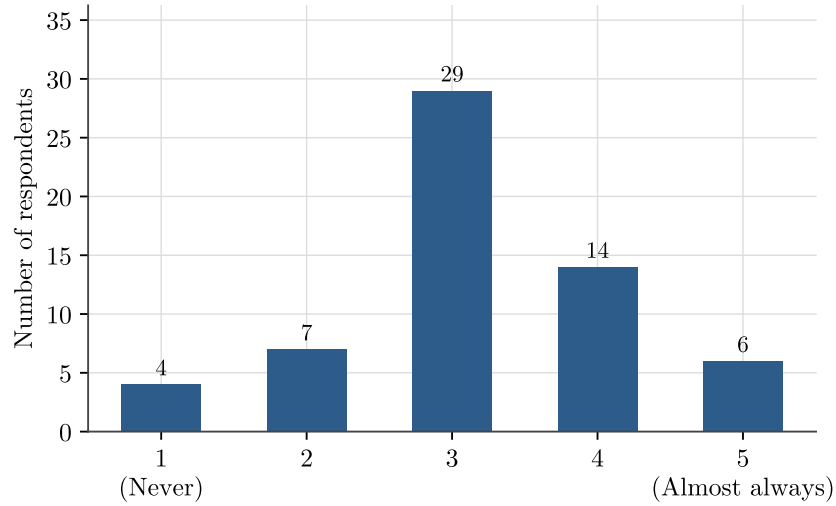


Figure 3.11: Tendency to change the planned activity when friends are present – Mean = 3.18, $SD = 1.00$, $N = 60$

This item is very different from D4 – instead of most answers clustering at the high end, nearly half the students (29/60, 48.3%) picked the middle option, “Sometimes.” Very few picked the extremes (“Never” or “Almost always”). This lines up well with what D1 found: friends being around often changes what a student does, but not every single time. So D5 confirms this is a real pattern, just not an automatic one – having friends nearby makes a student more likely to switch plans, but does not guarantee it. There is also a moderate link between D4 and D5 ($r = 0.49$): students who spend more time with friends overall are somewhat more likely to change their plans when friends show up, but the connection is not very strong.

3.5 Section E: Satisfaction with Campus Facilities

Item	Statement	Mean	SD
E1	Overall, I am satisfied with the campus spaces available to me during free periods.	3.07	1.03
E2	There is sufficient seating available on campus during peak free period times.	2.46	1.13
E3	Campus spaces adequately support the range of activities I want to do during free periods.	2.85	1.00

Table 3.3: Section E: Likert item means and standard deviations (1–5 scale, $N = 60$)

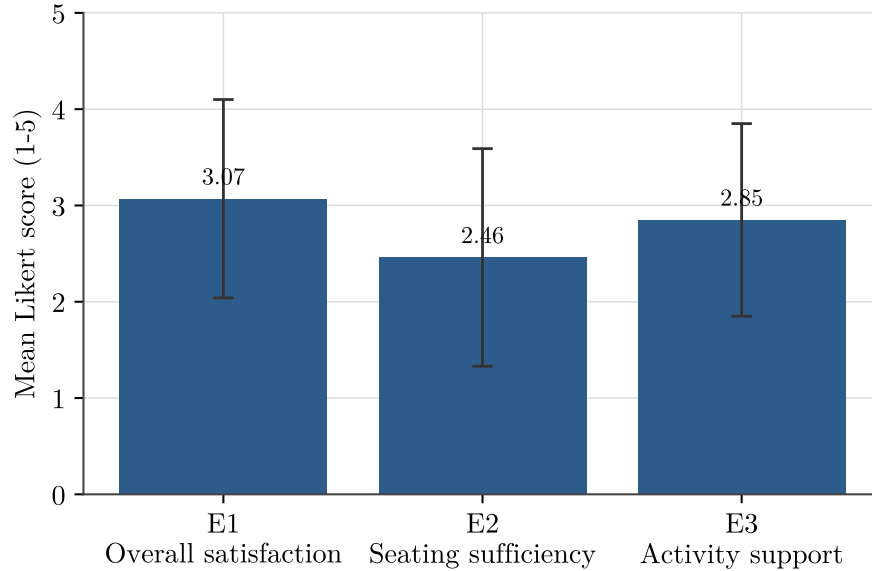


Figure 3.12: Satisfaction with campus facilities for free periods (error bars = 1 SD)

The three satisfaction items show a consistent pattern. Seating sufficiency (E2) received the lowest mean score (2.46), making it the weakest-rated item in the survey. More than half of the respondents (33 of 60, 55%) disagreed that there is enough seating during busy periods, while only 11 (18.3%) agreed. This supports the D1 interviews, where overcrowding was identified as a major concern, and matches the findings in Section 2.2, where overcrowding was the most common reason for avoiding the Canteen and the Open Area.

Activity support (E3) had a mean score of 2.85, indicating that students were slightly dissatisfied with how well campus spaces support different free-period activities. This is consistent with the D1 finding that some students felt the available spaces did not meet their activity needs.

Overall satisfaction (E1) had a mean score of 3.07, which is close to the neutral point. This suggests that students are not generally dissatisfied with campus spaces as a whole. Instead, their concerns are mainly related to specific issues, particularly limited seating and the suitability of spaces for different activities.

4 Cross-Cutting Patterns: Basic Comparisons

Beyond single-variable frequencies, the D2 objectives specifically call for identifying “the relationship between the length of a free period and the activity students engage in.” This chapter presents four basic cross-tabulations that connect background variables to behaviour.

4.1 Free Period Duration vs Primary Activity and Location

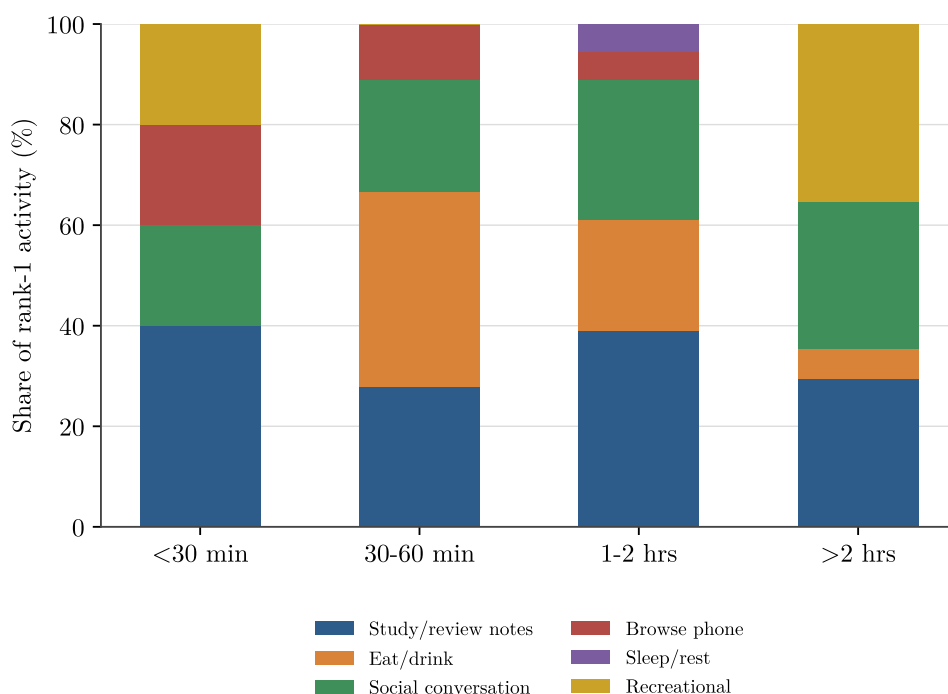


Figure 4.1: Primary (rank-1) free-period activity, by typical free period duration

Eating, drinking, and social conversation are relatively more common as the primary activity for shorter free periods, while studying/reviewing notes and recreational activity become comparatively more prominent as free periods lengthen. Phone browsing as a primary activity is concentrated in the shorter duration bands and essentially disappears as a top-ranked activity for periods over two hours. This is a strong, direct confirmation of the D1 cross-location insight that “free period duration shapes both location and activity type”.

4.2 Free Period Duration vs Influence of Next-Lecture Proximity

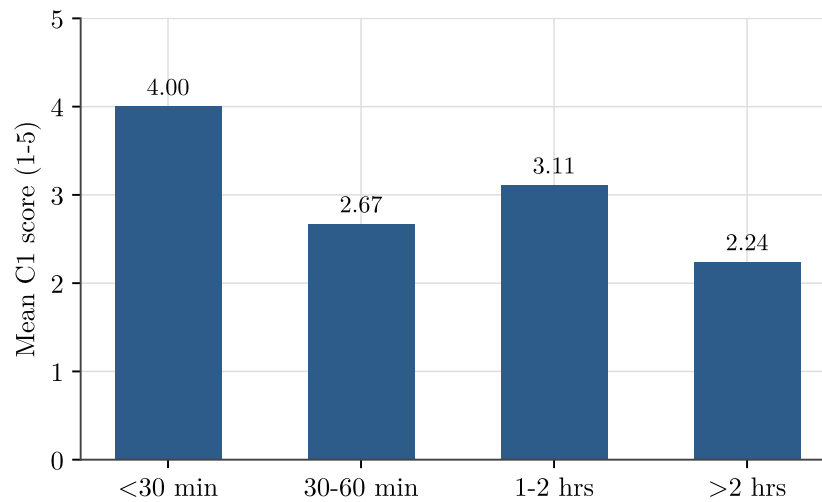


Figure 4.2: Mean C1 score (influence of next lecture's location), by free period duration

- Students with less than 30 minutes of free time gave an average score of 4.00, suggesting that distance to the next lecture hall strongly affects location selection.
- For students with longer free periods, the average score falls to 2.24.

This suggests that students with very short free periods tend to select locations near their lecture halls, since it is easier to get there in time. But for students with longer free periods, they select the most suitable place based on interest and comfort rather than distance to the next lecture. This gives a clear explanation for the D2 objective of measuring how the location of the next lecture influences a student's choice of free-period location.

4.3 Year of Study vs Overall Satisfaction

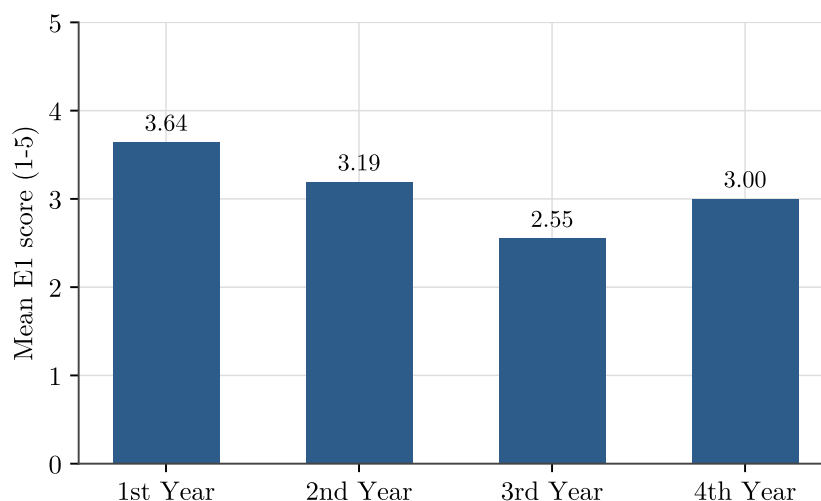


Figure 4.3: Mean overall satisfaction (E1), by year of study

An interesting result is that first-year students reported the highest satisfaction, with an average of 3.64, whereas third-year students reported the lowest satisfaction, with an average score of 2.55. This result is unexpected because first-year students have limited access to some locations, such as the Dewale and Common Room, and fourth-year students likewise make little use of most of those locations. The findings suggest that limited access is not the main driver of dissatisfaction. Instead, problems such as inadequate facilities, seating shortages, and overcrowding appear to be the larger factors. This pattern was not clear during the D1 qualitative study and should be explored further during the D4 phase.

4.4 Social Context vs Phone Browsing

Section D4 (frequency of spending free periods with friends, mean 3.88) and D2 (frequency of passive phone browsing, mean 3.38) show only a weak positive correlation ($r = 0.16$) across the sample. This is a more modest relationship than the strong solo/group contrast in phone use that D1 observation suggested – students on their own were seen mostly on their phones, while students in groups used phones as shared social objects. The survey-level frequency measures do not, on their own, capture the qualitative distinction D1 drew between passive individual phone use and active social phone use within groups. This is noted as a genuine gap between what the D1 fieldwork observed and what the D2 instrument was able to measure, to be addressed narratively in the D4 report rather than through additional D3 statistics.

5 Linking Quantitative Findings to the D1 Themes and D2 Objectives

5.1 Revisiting the D2 Research Objectives

Table 5.1 maps each of the six D2 research objectives to the D3 evidence that addresses it, showing that the quantitative phase answers every objective set out at the start of the study.

D2 Research Objective	D3 Evidence
Identify the most frequently used location	46.7% selected the Library as their first choice, making it the most frequently used location (Section 3.2)
Identify main activities and their relative frequency	Studying (31.7%), social conversation (25.0%), and eating (20%) are the most common activities (Section 3.4)
Measure how common passive phone browsing is	It appeared in the top-three choices of 31.7% of students but was the main activity for only 6.7%. Average frequency score was 3.38/5 (Section 3.4)
Measure influence of next lecture location	Mean C1 = 2.80/5 overall, but rises to 4.00 for free periods under 30 minutes (Chapter 4)
Assess satisfaction with existing facilities	All 3 satisfaction items scored around or below the midpoint; E2 (seating availability) got the lowest score (Section 3.5)
Identify the relationship between length of free period and activity	Confirmed that shorter periods are mainly used for eating, chatting, and phone browsing, whereas longer free periods are used for studying and recreation (Chapter 4)

Table 5.1: Mapping of D2 research objectives to D3 quantitative evidence

5.2 Revisiting the D1 Themes

Table 5.2 summarises, for each D1 cross-location insight, whether the D3 quantitative data agrees, partially agrees, or reveals a contradiction or new pattern.

D1 cross-location insight	Verdict	D3 Evidence
Location matches activity, not habit	Supported	C4 received a higher mean score of 4.07/5
Free-period duration shapes activity	Supported	There is a clear relationship between primary activity and free period duration (Figure 4.1)
Phone use depends on social context	Partially supported	Phone browsing is chosen by 31.7% of respondents as one of their top-3 activities, but its correlation with solo/group behaviour is weak ($r = 0.16$)
Solo vs group occupy spaces differently	Partially supported	Not directly separable at survey level using D4/D5 frequencies alone
Environmental conditions filter space use	Supported	C5 had the highest mean score (4.10/5)

Table 5.2: D1 qualitative themes against D3 quantitative evidence

Taken together, Tables 5.1 and 5.2 show that the D3 quantitative study successfully addressed all six research objectives defined in D2 while also testing the main themes identified during the D1 fieldwork. The survey strongly supports three key D1 findings: students choose spaces based on the activity they plan to do, environmental conditions influence space usage, and the length of a free period affects activity choice.

The other two D1 themes, phone use and solo/group behaviour, were also supported by the survey but with less detail than the field observations. This shows that the quantitative and qualitative phases complement each other. The survey provides statistical evidence for the observed patterns, while the fieldwork offers additional context and explains how and why those patterns occur. Together, these findings provide a strong foundation for the D4 report to further interpret and integrate the results.

6 Critical Analysis

6.1 Importance of Separating Access Restrictions from Personal Avoidance

One of the most important analytical decisions in this report was to analyse structural access restrictions separately from preference-based reasons for avoiding campus spaces (Section 2.2). If these had been combined, the Dewale and Common Room would have appeared to be avoided as often as the Canteen, even though the reasons are very different. Students mainly avoid the Canteen because of issues such as poor ventilation, unpleasant smells, overcrowding, and limited seating, which the university could improve. In contrast, the Dewale and Common Room are avoided mainly because some students are not allowed to access them due to university rules.

This clearly shows the difference between access restrictions and facility problems. It also supports previous research, which explains that the use of campus spaces is influenced not only by the physical environment but also by university rules and policies.

6.2 Limitations and Weaknesses of the Survey

- **Unclear “Common Room” wording:** The D0, D1, and D2 reports, as well as the D2/D3 survey, referred to the “Common Room” without specifying whether it meant the second-year or third-year common room. This report assumes that respondents referred to the common room for their own year group, an interpretation that is supported by the survey results but not explicitly confirmed by the questionnaire. A year-qualified version of this question would have been clearer.
- **Small number of fourth-year students:** Only 4 fourth-year responses were collected, consistent with D1’s finding that this group is rarely present at the five surveyed locations, because they are mostly in research labs working on their final-year projects. Therefore, any year-based comparison involving fourth-years should be read as indicative only.
- **Self-reported responses:** The survey relies on students reporting their own behaviour. Even though the phone-browsing question was improved during the D2 pilot study, many students still selected neutral answers. This suggests that some students may have answered in a socially acceptable way instead of describing their actual behaviour.
- **Data collected during a single time period:** The survey was conducted over nine days in July, when there were no examinations and most students did not have assignments due. Student behaviour could vary during exam periods, so the findings

may not represent student behaviour throughout the entire academic year.

- **Weak measurable link between social context and phone use:** In Chapter 4, only a weak link was found between how often a student is with friends (D4) and how often they browse their phone (D2). This does not mean that the D1 finding – that students on their own tend to browse passively, while students in groups often use their phones socially – is wrong. A single rating-scale question cannot separate these two different types of phone use; it only asks “how often,” not “in what way” or “with what purpose.” This is a sign that this particular question was not detailed enough to capture the distinction that D1’s direct observation could see.

6.3 Reflection on the D1–D2–D3 Pipeline

The D3 survey closely confirmed the D1 findings when the D2 questionnaire included questions that were directly based on D1 observations, such as environmental conditions, activity-location fit, and Canteen odour. However, when the questionnaire only partly covered a D1 insight (such as solo/group phone use) or did not include a specific observation that the D1 interviews revealed (such as the lack of a space for unwell students), the survey could not fully examine those findings.

This highlights an important lesson for the D4 report: not all qualitative findings can be measured effectively using survey questions. Some themes are better understood through qualitative evidence and should remain part of the final integrated discussion.

7 Conclusion and Future Works

7.1 Conclusion

This D3 report has shown that the full-scale survey of 60 students strongly supports several key findings from D1, particularly that free-period length shapes both activity and location choice, and that environmental factors such as crowding and smell drive avoidance of the Canteen and Open Area. It also uncovered new insights, such as the genuine access gap facing first- and fourth-year students, and the unexpected finding that third-years, despite having full access to campus spaces, report the lowest satisfaction. Together, these quantitative results confirm, refine, and in some cases challenge the qualitative themes identified earlier, providing a solid evidence base to be integrated with the D1 findings in the final D4 report.

7.2 Future Works

D4 – Final Research Report: The final stage of this research will integrate the qualitative findings of D1 with the quantitative findings presented in this D3 report into a single, comprehensive analysis of student behaviour during free periods at UCSC. The D4 report will explicitly reconcile the confirmations, partial confirmations, and gaps identified in Chapter 5, revisit the two D1 themes that this survey did not directly measure using the original interview evidence, and critically discuss the implications of the equity finding around restricted-access spaces for campus space planning. The strengths and limitations of the overall mixed-methods design, as reflected on in Chapter 6, will also be discussed in full in the D4 submission.